2. b) $P_{(B|\bar{T},\Delta)} = \frac{P_{(B,\bar{T},\Delta)}}{P(\bar{T},\Delta)}$

$P_{(B,\bar{T},\Delta)} = P_{(B)} P_{(\bar{T}|B)} P_{(\Delta|B,\bar{T})}$

$P_{(B|\bar{T},\Delta)} = \frac{P_{(B)} \cdot P_{(\bar{T}|B)} \cdot P_{(\Delta|B,\bar{T})}}{\sum_{b \in \{0,1\}} P_{(B=b)} P_{(\bar{T}=1|B=b)} P_{(\Delta=1|B=b,\bar{T}=1)}}$

$P_{(B|\bar{T},\Delta)} = \frac{0,1 \cdot 0,8 \cdot 0,9}{0,1 \cdot 0,8 \cdot 0,9 + 0,9 \cdot 0,3 \cdot 0,5} = \frac{0,072}{0,072 + 0,135} =$

$= \frac{0,072}{0,207} = 0,3478$

c) $P_{(X=1|B=0)} = \frac{P_{(B=0|X=1)} \cdot P_{(X=1)}}{P_{(B=0)}}$

$P_{(X=1|B=0)} = \frac{P_{(X=1,B=0)}}{P_{(B=0)}} = \frac{0,1 \cdot 0,9}{0,9} = 0,1$

Scanned with CamScanner